	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 5

Lab Title: **Design and Configuration of Half and Full Adder**

Student Name: **Muammad Waqas** **Reg. No:** **240756**

Objective: To experimentally verify the operation of basic adders,various configurations of adders.

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

TASK1:

Design a Half Adder circuit and complete the truth table.

Steps:

1. Complete the truth table.
2. Write the logic equations for Sum and Carry.
3. Draw the complete logic diagram for the Half Adder.
4. Verify your results practically.

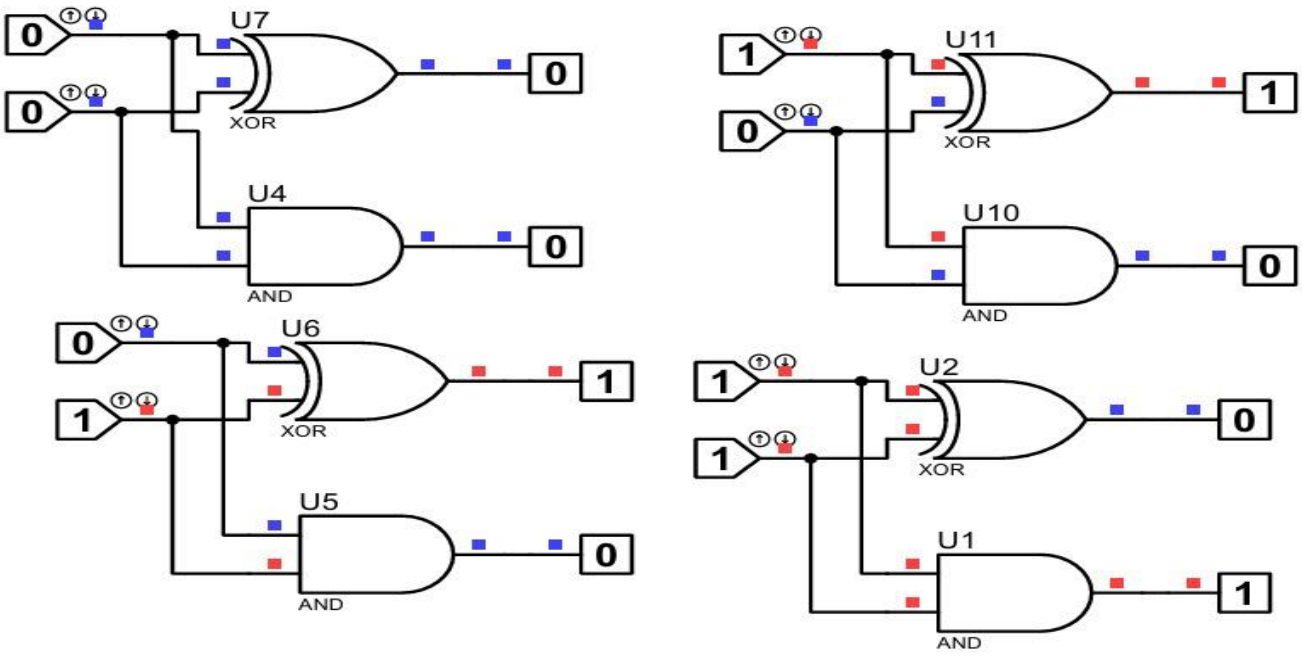
A	B	CO	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

2 Logic Equation for sum and carry

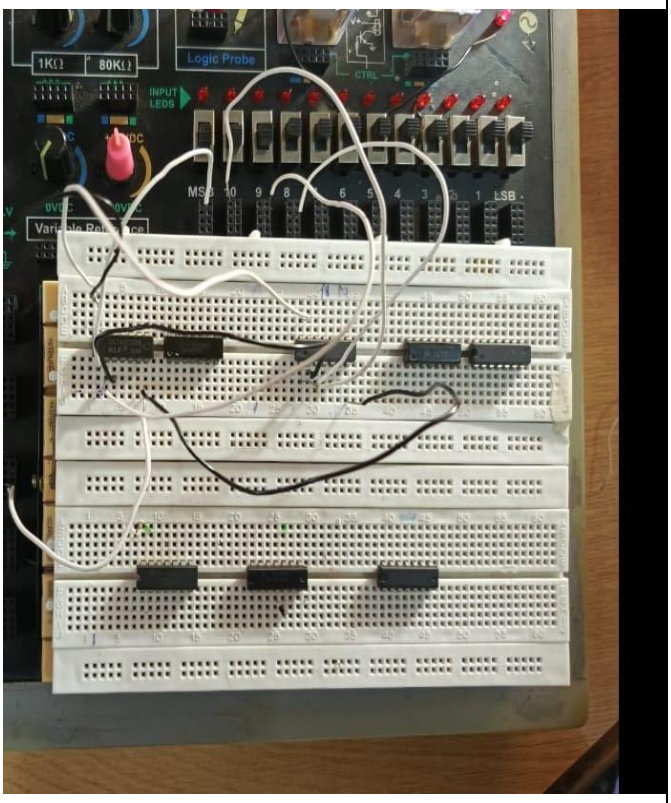
SUM	A'B+AB'
CARRY	A.B

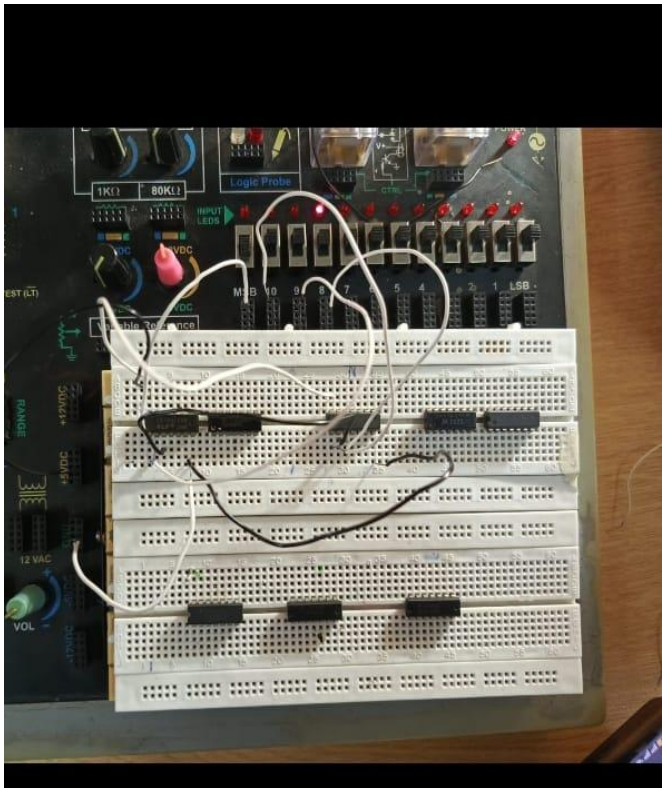
$$\textbf{Sum} = A \oplus B$$
$$\textbf{Carry} = A \cdot B$$

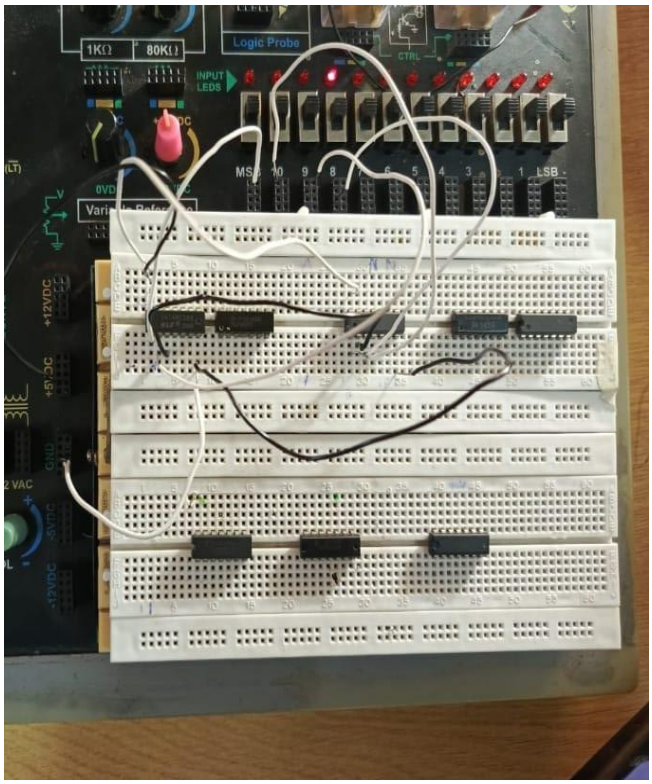
3. Complete diagram of Half adder:

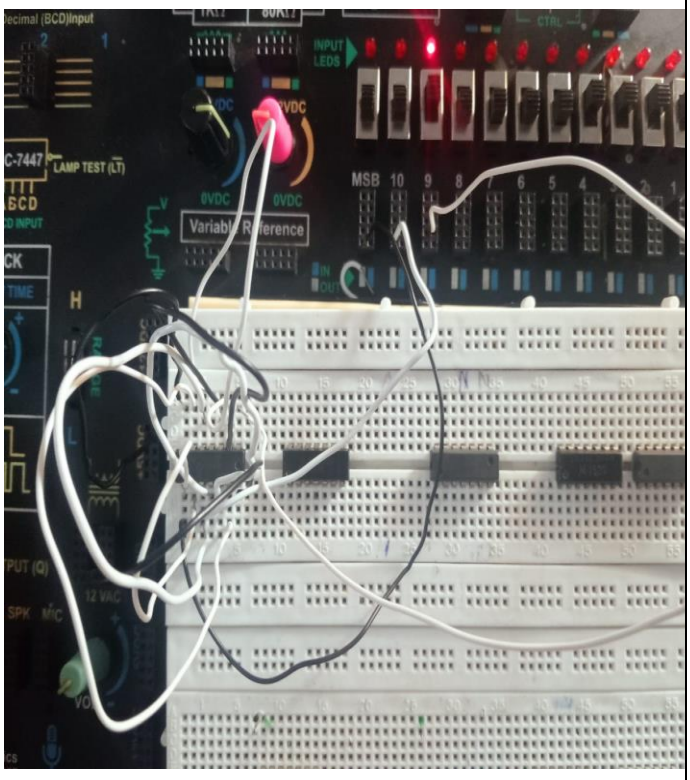


4 Practical simulation:







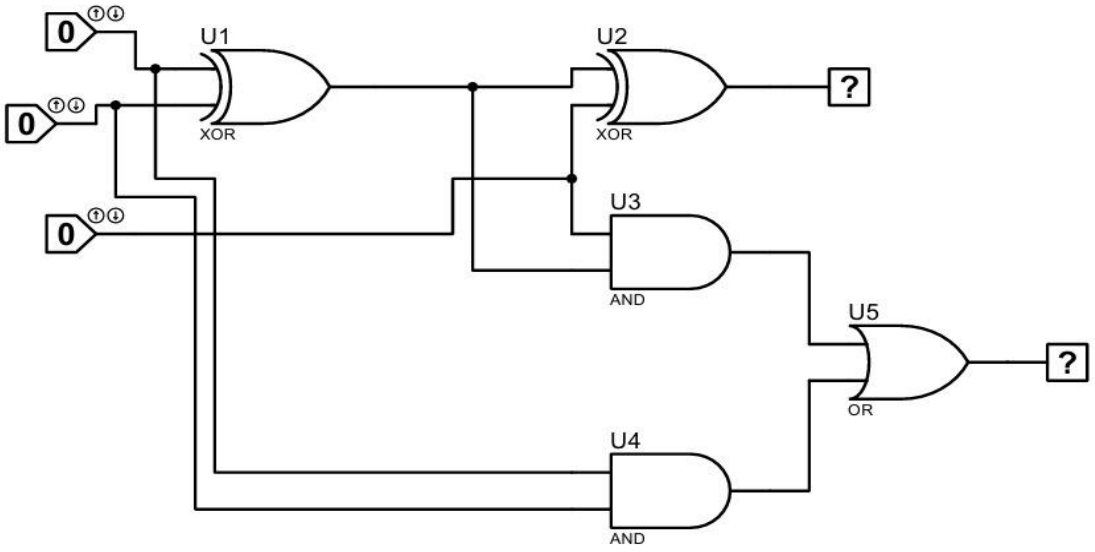


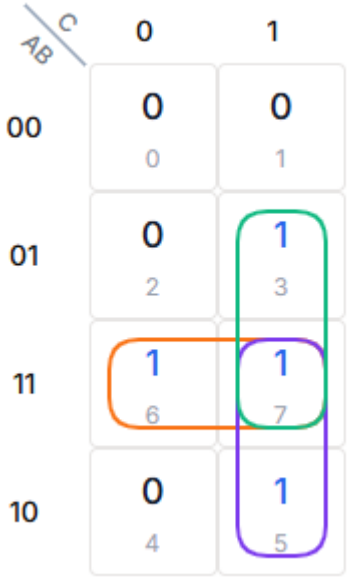
Task 2:

Design a Full Adder circuit and complete the truth table.

Steps:

1. Draw the complete logic diagram.
2. Write the logic equations for Sum and Carry out.
3. Complete the Truth Table.
4. Implement the circuit on trainer and verify your results practically.



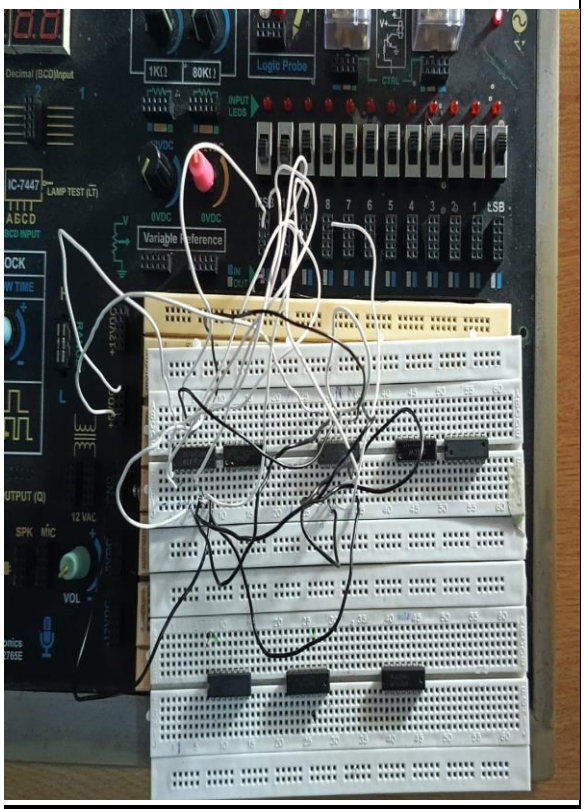
K-Map for Sum	K-map for Carry
	
Simplified expression: A'B'C+A'BC'+AB'C'+ABC	**Simplified expression:** AB+AC+BC

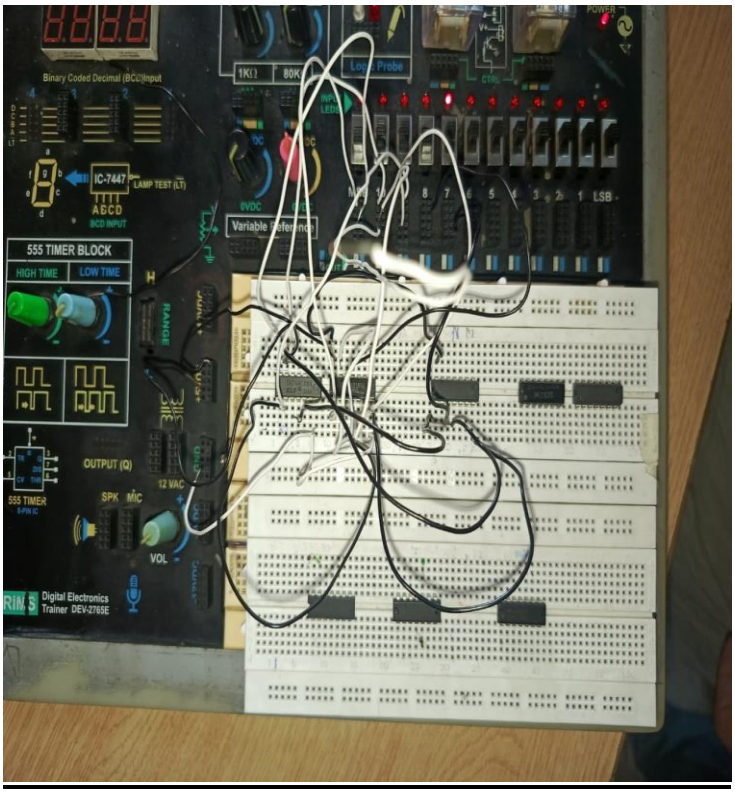
SUM	CARRY
The sum: S = X'Y'C_in + X'YC_in' + XY'C_in' + XYC_in = C_in'(X'Y + XY') + C_in(X'Y' + XY) = C_in'(X ⊕ Y) + C_in(X ⊕ Y)' **S = C_in ⊕ (X ⊕ Y)**	The carry output: C_out = X'YC_in + XY'C_in + XY C_in' + XYC_in = C_in(X'Y + XY') + XY(C_in + C_in') **C_out = C_in·(X ⊕ Y) + XY**

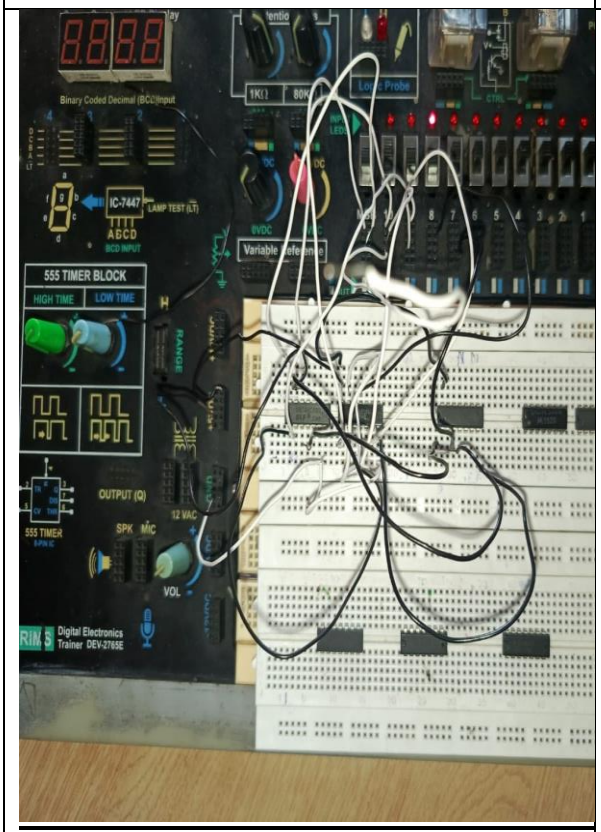
TRUTH TABLE:

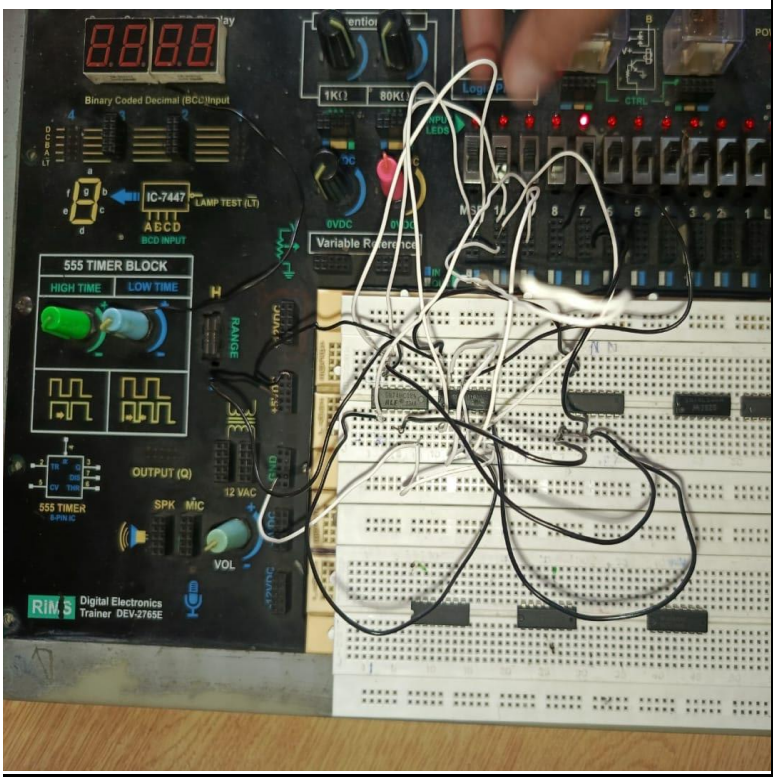
ROW	X	Y	Cin	S	Cout
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

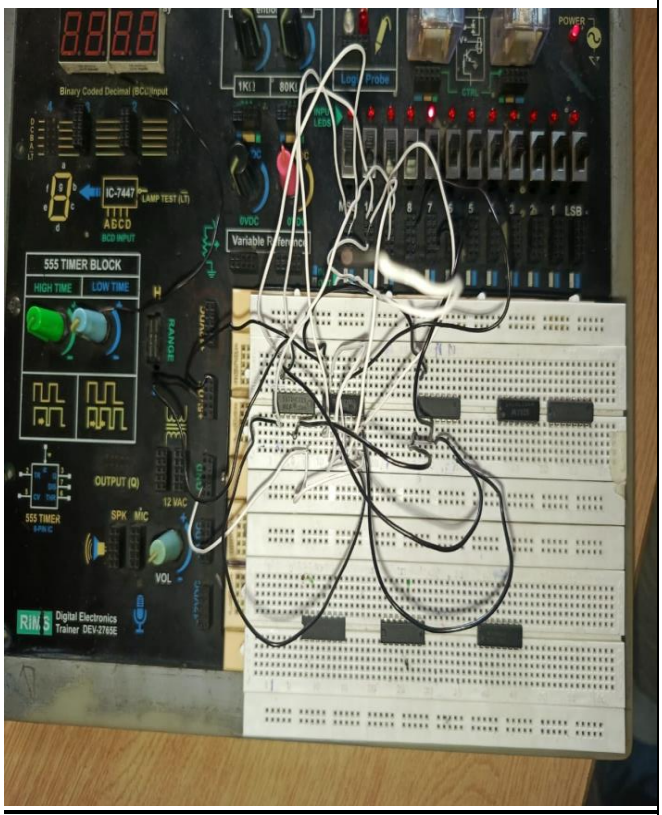
Practical Simulation:

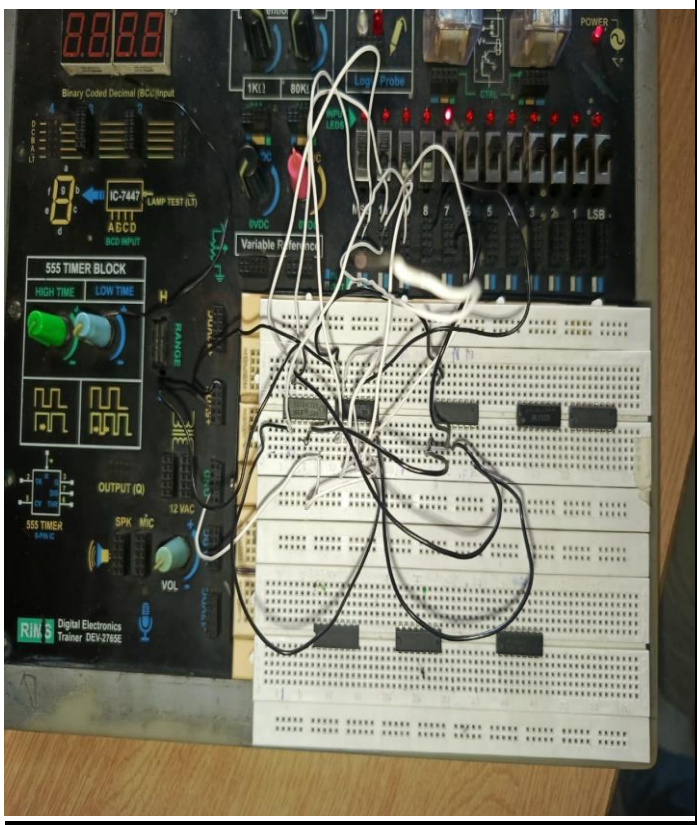


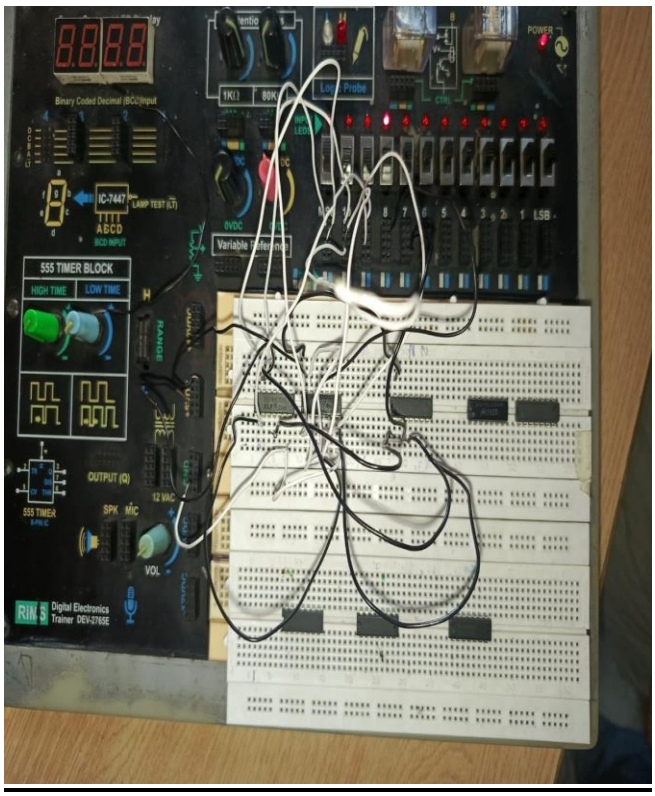


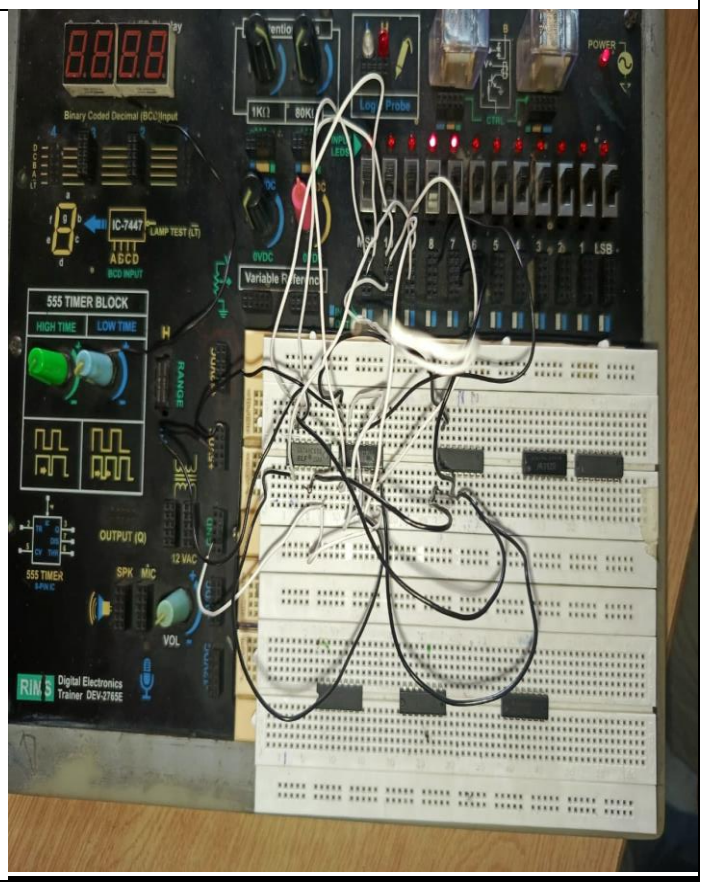





Binary Coded Decimal (BCD) Input
IC-7447
LAMP TEST (LT)
ABCD
BCD INPUT
555 TIMER BLOCK
HIGH TIME
LOW TIME
RANGE
OUTPUT (Q)
555 TIMER
SPK
MIC
VOL
12 VAC
0VDC
Variable Reference
Digital Electronics
Trainer DEV-2765E


Binary Coded Decimal (BCD) Input
IC-7447
LAMP TEST (LT)
ABCD
BCD INPUT
555 TIMER BLOCK
HIGH TIME
LOW TIME
RANGE
OUTPUT (Q)
555 TIMER
SPK
MIC
VOL
12 VAC
0VDC
Variable Reference
Digital Electronics
Trainer DEV-2765E


Binary Coded Decimal (BCD) Input
IC-7447
LAMP TEST (LT)
ABCD
BCD INPUT
555 TIMER BLOCK
HIGH TIME
LOW TIME
RANGE
OUTPUT (Q)
555 TIMER
SPK
MIC
VOL
12 VAC
0VDC
Variable Reference
Digital Electronics
Trainer DEV-2765E


Binary Coded Decimal (BCD) Input
IC-7447
LAMP TEST (LT)
ABCD
BCD INPUT
555 TIMER BLOCK
HIGH TIME
LOW TIME
RANGE
OUTPUT (Q)
555 TIMER
SPK
MIC
VOL
12 VAC
0VDC
Variable Reference
Digital Electronics
Trainer DEV-2765E

Conclusion:

A Half Adder and a Full Adder are basic building blocks of digital arithmetic circuits used for binary addition. The Half Adder adds two single-bit binary numbers and produces two outputs — Sum and Carry. However, it cannot handle a carry input from a previous stage. The Full Adder overcomes this limitation by including a third input, the Carry-in, allowing it to add three binary bits. It produces a Sum and a Carry-out.
In summary, the Half Adder is used for simple one-bit additions, while the Full Adder is used in multi-bit binary addition systems such as adders, ALUs, and processors by connecting multiple full adders in series to perform complex arithmetic operations efficiently.